

Number of registers

Did you know that a graphics card can have as many as 3072 processors and each processor can have up to 255 registers? While the number of processors is probably enough to amaze you, what is more amazing is the number of registers available to the processors! If you compare this number to what's in a normal PC processor, it's huge. Most x86 processors have only about 20 registers to work with. If you've ever worked with assembly, you'd appreciate the luxury of being able to work with 255 registers at one time. If you haven't, take our word for it, it feels spacious. Very spacious.

For those of you not familiar with the term 'register': A register is a very small and extremely high-speed memory that the processor can directly access, and requires exactly one cycle for the processor to retrieve data from it. Thus, obviously, the more registers your program uses, the faster it runs. Also, a program using the registers will run faster than one using the system RAM. The truth is, RAM can be hundreds of times slower than accessing data from registers.

Let's recap and take a look at the list of challenges one could face when designing or selecting a processor for an embedded system:

- 1 Power Consumption
- 2 Processor type – RISC / CISC
- 3 Word Length
- 4 Memory Addressability
- 5 Register count

And this is just for the processor!

Depending on the specific use-case, the right mix of the above can help get the best results. Again, the processor alone is like a severed head; other components are equally vital – each of them with their own design challenges.

Design challenges when building RAM

Without relying on your memory, you'd get very little done. The same applies to the electronics world. Every piece of equipment that can calculate or give commands should be able to store some data in the system's primary memory, also known as RAM. You probably already know that RAM is a fast but volatile memory closer to the processor, and is much faster than secondary storage in terms of the time taken to access it.

In most (if not all) cases, embedded systems have the main memory embedded onto the main board itself. The performance requirements are strict and real-time and there's no scope for altering or replacing the com-